MATTHEW DE LANNOY

Robotics Engineer

matthewdelan@gmail.com | +31 6 45035492 | Delft, The Netherlands | matt-rbt

PROJECTS

Cora: Collaborative Robot Platform | Mar 2025 – Present
- Lead ME/EE/SW engineer on open-source cobot (1.0 m reach, 3 kg payload); led cross-disciplinary team of 5
- Architected distributed control system using ROS2, Ethernet TCP/IP, CAN bus motor controllers, and Python SDK
- Designed modular joint mechanisms for scalable robot architecture

You vs Cobot: Chess Application | Mar 2025 – Present
- Autonomous chess system with vision-based board recognition
- CNN-based move selection showcasing full-stack robot integration

Cavolab: Automated Cable Cleaning (Prysmian Minor Project) | Sept 2025 – Feb 2026
- University minor project in collaboration with Prysmian Group
- Led mechanical design of gantry system and pneumatic cleaning structures
- Delivered prototype validation and technical documentation for potential industry application

Atlas: 6-DOF Desktop Arm | Sept 2024 – Mar 2025
- Designed articulated robotic arm (500 mm reach, 1 kg payload)
- Developed embedded control system on ESP32
- Implemented ROS 2 control and MoveIt 2 stack
- Early prototype contributing to Cora platform development

PROFESSIONAL EXPERIENCE

Teaching Assistant – VideoFastLane
TU Delft - TLS

Feb 2026 – Apr 2026

2D animation support, student guidance, and course communication

Teaching Assistant – Robotics
TU Delft - ME

Feb 2026 – Apr 2026

Lab instruction, assessment, and robotics fundamentals support

Teaching Assistant – CAD
TU Delft - ME

Nov 2023 – Feb 2026

SolidWorks instruction and CAD mentorship

Student Mentor
TU Delft - ME

Sept 2024 – Feb 2025

Academic and professional guidance for undergraduate students

EDUCATION

B.Sc. Mechanical Engineering
TU Delft

Sept 2022 – June 2026

Minor in Robotics
TU Delft

Sept 2025 – Jan 2026

EXTRACURRICULAR

Workshop Committee | RSA Delft
Facility management and technical workshop coordination

Competitions
- **DCSC Hackathon | DCSC - TU Delft** Ping pong ball thrower control system using open source Aeroshield
- **Underground Hackathon | Samxl** Remote controlled duct navigating and inspection robot
- **Renewable Energy Hackathon | Samxl** Turbine blade inspection robot for offshore wind farms
- **Boskalis Hackathon | Boskalis** Autonomous seagrass planting robot for coastal restoration

TECHNICAL SKILLS

Robotics & Embedded Systems

ROS 2, Ethernet TCP/IP, CAN Bus, Raspberry Pi, RP2040, Real-time Control

Software

Python, C++, Git, Firmware Development

Design & CAD

SolidWorks, Onshape, FEA, 3D Printing, CNC Machining

Electronics

KiCAD, Microcontrollers, Motor Control, Sensor Integration

Professional

Agile/Scrum, Cross-team Communication